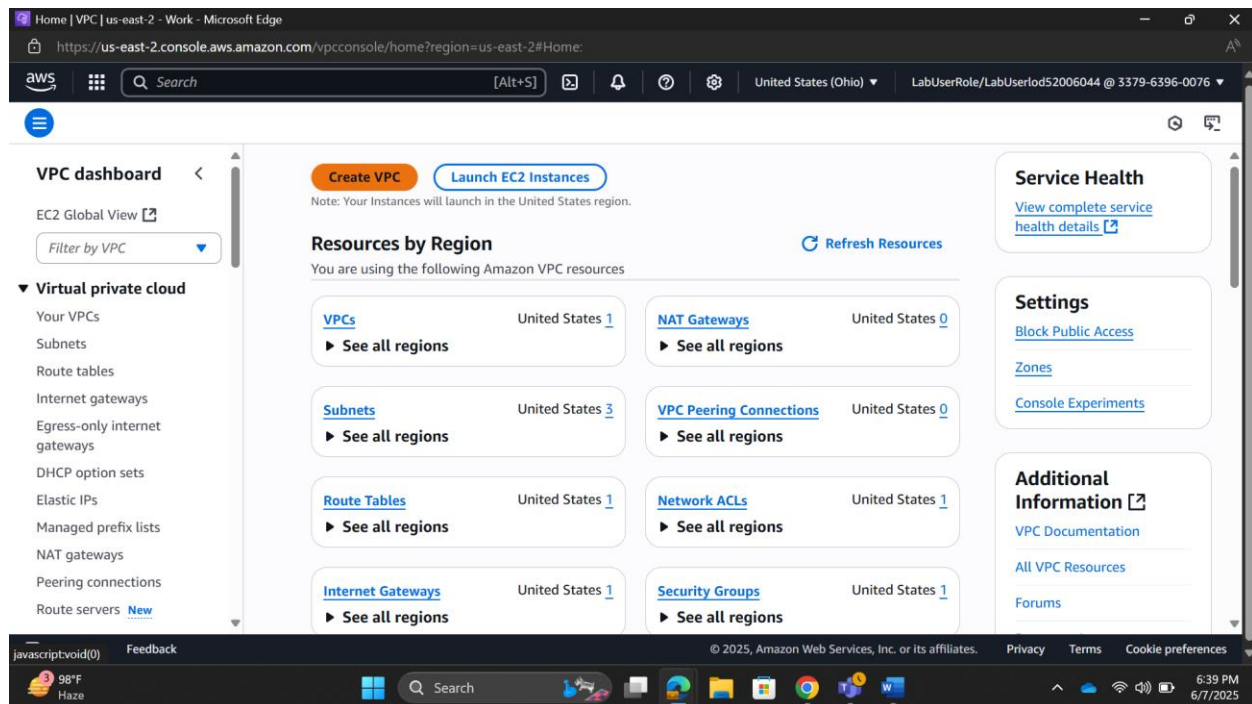


Lab 2 Build a Network by Using the VPC Wizard

In this challenge, you will create a custom VPC with public and private subnets, and an Internet and NAT Gateway. You will then associate route tables with the correct subnets. Note: Once you begin a challenge you will not be able to pause, save, or return to your progress. Please ensure you have set aside enough time to complete the challenge before you start.

Task 1: Create a base network by using the VPC Wizard

VPC -> Create VPC



The screenshot shows the AWS VPC console interface. The top navigation bar includes the AWS logo, a search bar, and the current region (United States (Ohio)). The left sidebar contains the 'VPC dashboard' and a list of VPC resources. The main content area is titled 'Resources by Region' and shows a grid of resource cards for the United States (Ohio) region. Each card displays the resource name, a link to 'See all regions', and the count of resources in that region. The resources listed are:

- VPCs: 1
- NAT Gateways: 0
- Subnets: 3
- VPC Peering Connections: 0
- Route Tables: 1
- Network ACLs: 1
- Internet Gateways: 1
- Security Groups: 1

The right sidebar contains sections for 'Service Health', 'Settings', and 'Additional Information'. The bottom of the screen shows the Windows taskbar with the time 6:39 PM on 6/7/2025.

VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#CreateVpc:createMode=vpcWithResources

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Your VPCs > Create VPC

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate
Lab

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IPs
CIDR block size must be between /16 and /28.

Preview

VPC [Show details](#)
Your AWS virtual network

Lab-vpc

Subnets (4)
Subnets within this VPC

us-east-2a

- Lab-subnet-public1-us-east-2a
- Lab-subnet-private1-us-east-2a

us-east-2b

- Lab-subnet-public2-us-east-2b
- Lab-subnet-private2-us-east-2b

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VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#CreateVpc:createMode=vpcWithResources

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Your VPCs > Create VPC

Number of Availability Zones (AZs) [Info](#)
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1 | 2 | 3

► Customize AZs

Number of public subnets [Info](#)
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0 | 1

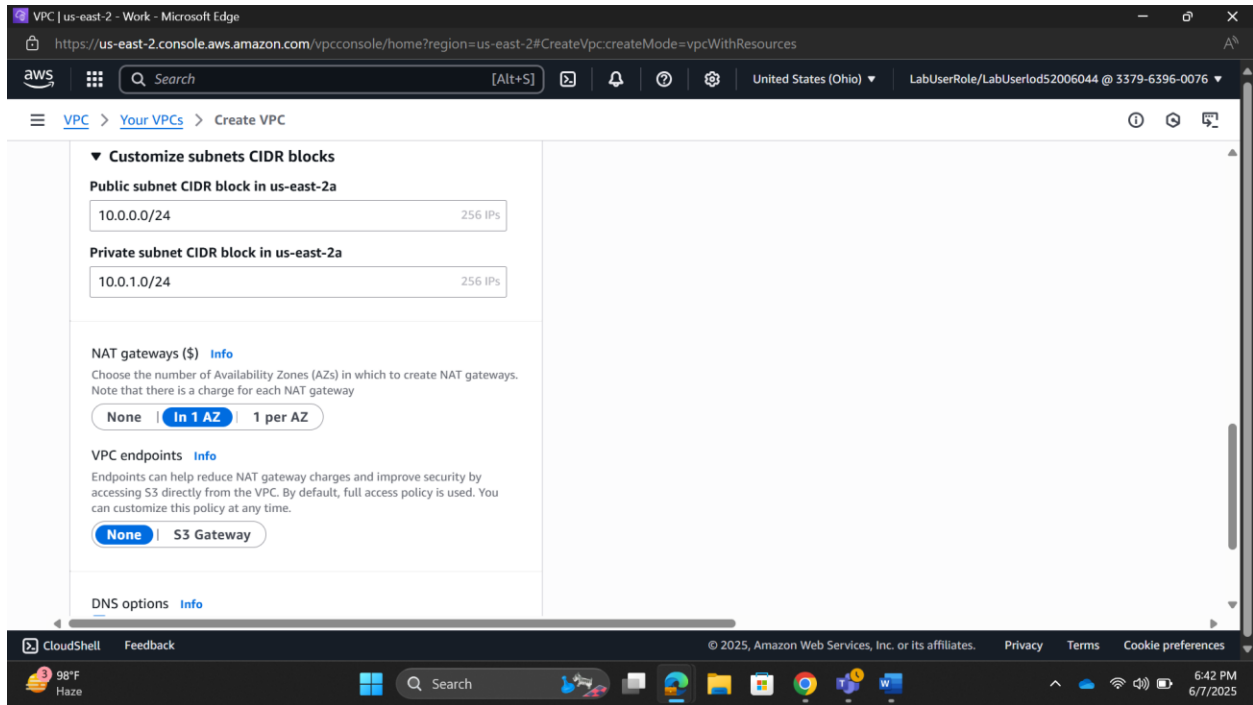
Number of private subnets [Info](#)
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 | 1 | 2

► Customize subnets CIDR blocks

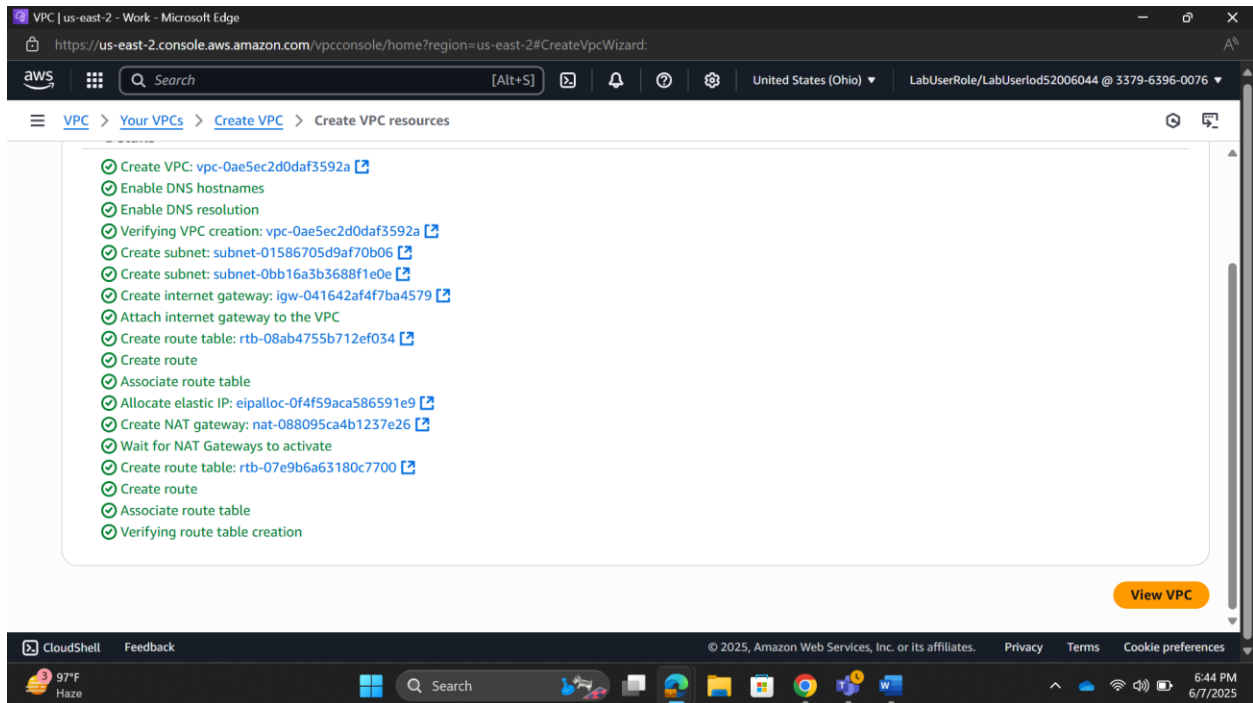
NAT gateways (\$) [Info](#)
Choose the number of Availability Zones (AZs) in which to create NAT gateways.

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Click create VPC

Click view VPC



Task 2: Create additional subnets

VPC -> Virtual private cloud -> subnets -> create subnet

The screenshot shows the AWS Management Console interface for the 'us-east-2' region. The main content area is titled 'Subnets (3)' and displays a table of existing subnets. The table has columns for Name, Subnet ID, State, and VPC. All three subnets are in an 'Available' state and are associated with the VPC 'vpc-0cc60d13ebe8fa397'. The left sidebar shows the 'Virtual private cloud' section with 'Subnets' selected. The bottom of the screen shows a Windows taskbar with the date 6/7/2025 and time 6:46 PM.

Name	Subnet ID	State	VPC
-	subnet-0d799df0cbab495eb	Available	vpc-0cc60d13ebe8fa397
-	subnet-019c239599cc283ba	Available	vpc-0cc60d13ebe8fa397
-	subnet-03f7c3ac33cea9ba7	Available	vpc-0cc60d13ebe8fa397

VPC | us-east-2 - Work - Microsoft Edge
https://us-east-2.console.aws.amazon.com/vpccconsole/home?region=us-east-2#CreateSubnet:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Subnets > Create subnet

Create subnet [Info](#)

VPC
VPC ID
Create subnets in this VPC.
vpc-0ae5ec2d0daf3592a (Lab-vpc)

Associated VPC CIDRs
IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1
Subnet name
Create a tag with a key of 'Name' and a value that you specify.

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VPC | us-east-2 - Work - Microsoft Edge
https://us-east-2.console.aws.amazon.com/vpccconsole/home?region=us-east-2#CreateSubnet:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public Subnet 2
The name can be up to 256 characters long.

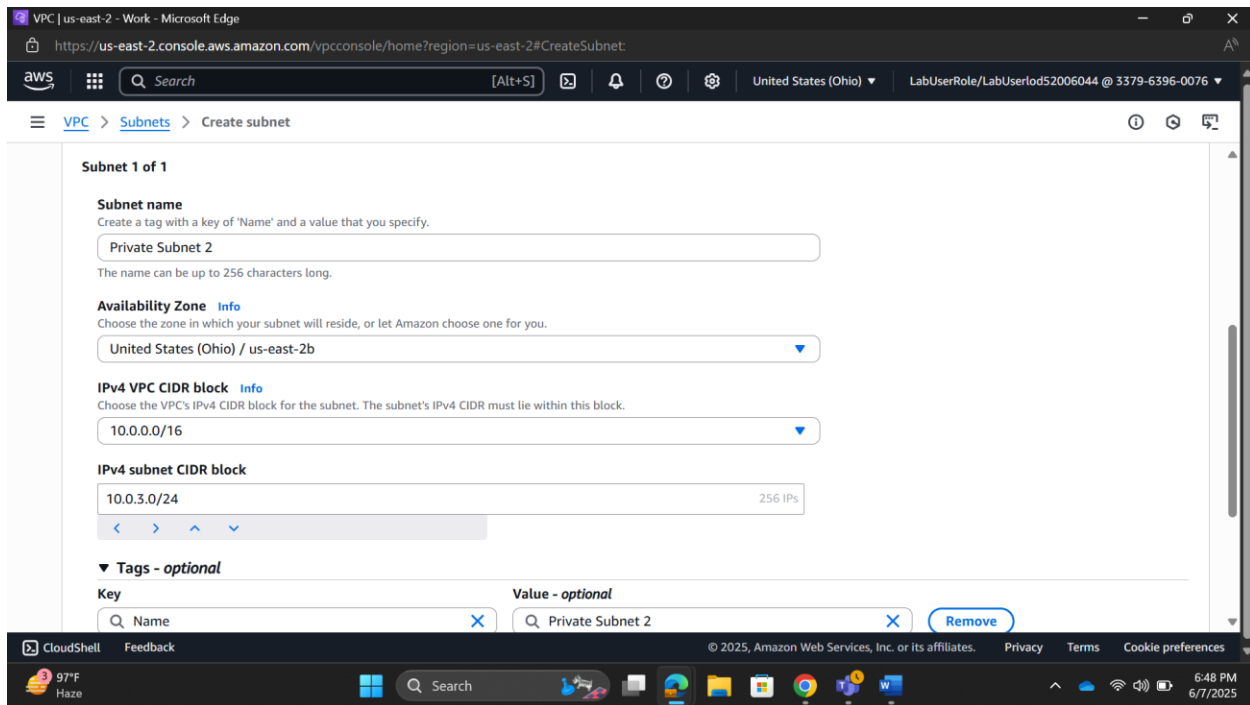
Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (Ohio) / us-east-2b

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.2.0/24 256 IPs
< > ^ v

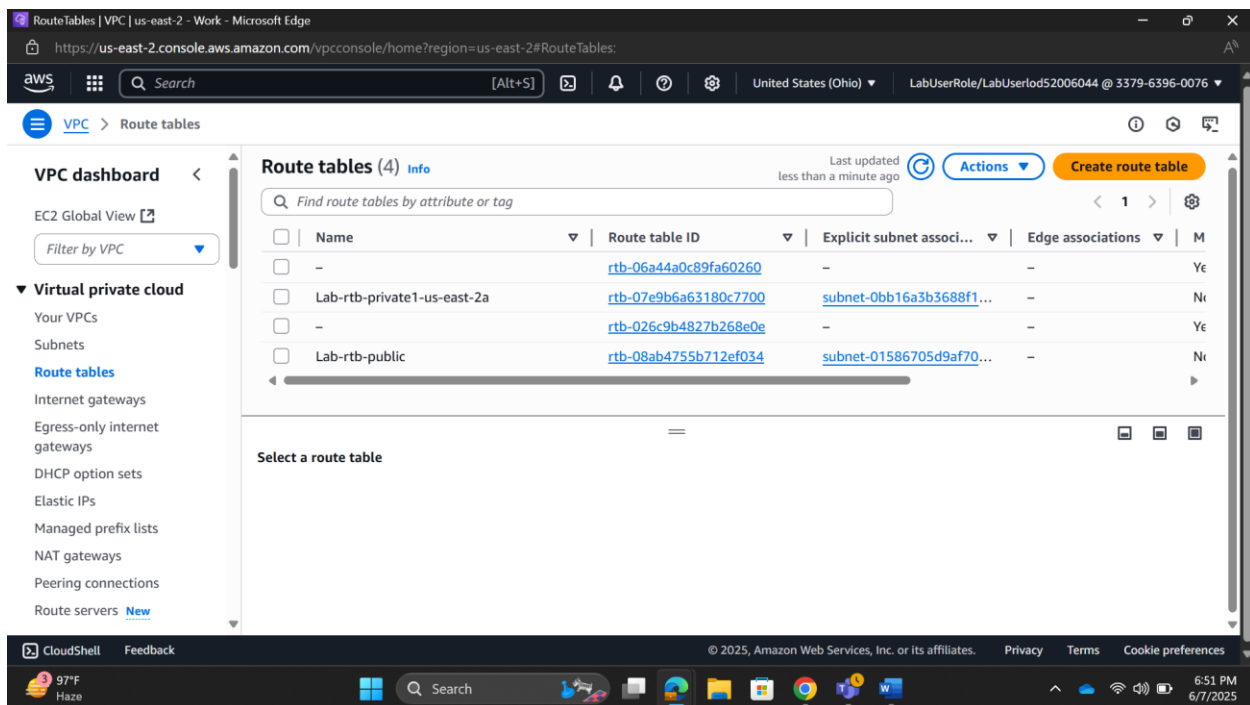
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Create another private subnet



Task 3: Configure route tables

VPC -> Virtual private cloud -> Route tables



RouteTables | VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTables:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Route tables

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Route tables (1/4) Info

Last updated 1 minute ago

Actions Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations
-	rtb-06a44a0c89fa60260	-	-
Lab-rtb-private1-us-east-2a	rtb-07e9b6a63180c7700	subnet-0bb16a3b3688f1...	-
-	rtb-026c9b4827b268e0e	-	-

rtb-07e9b6a63180c7700 / Lab-rtb-private1-us-east-2a

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Both Edit routes

Destination	Target	Status	Propagated
0.0.0.0/0	nat-088095ca4b1237e26	Active	No
10.0.0.0/16	local	Active	No

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RouteTables | VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTables:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Route tables

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Route tables (1/4) Info

Last updated 2 minutes ago

Actions Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations
-	rtb-06a44a0c89fa60260	-	-
Lab-rtb-private1-us-east-2a	rtb-07e9b6a63180c7700	subnet-0bb16a3b3688f1...	-
-	rtb-026c9b4827b268e0e	-	-

rtb-07e9b6a63180c7700 / Lab-rtb-private1-us-east-2a

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Both Edit routes

Destination	Target	Status	Propagated
0.0.0.0/0	nat-088095ca4b1237e26	Active	No
10.0.0.0/16	local	Active	No

Edit Name

Private Route Table

Cancel Save

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RouteTables | VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTables:

aws

Search [Alt+S]

United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Route tables

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Route tables (1/4) Info

Last updated 1 minute ago

Actions Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associations
-	rtb-06a44a0c89fa60260	-
☒ Private Route Table	rtb-07e9b6a63180c7700	subnet-0bb16a3...
-	rtb-026c9b4827b268e0e	-

rtb-07e9b6a63180c7700 / Private Route Table

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated
0.0.0.0/0	nat-088095ca4b1237e26	Active	No
10.0.0.0/16	local	Active	No

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RouteTables | VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTables:

aws

Search [Alt+S]

United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Route tables

VPC dashboard

EC2 Global View

Filter by VPC

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Egress-only internet gateways

DHCP option sets

Elastic IPs

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NAT gateways

Peering connections

Route servers

Route tables (1/4) Info

Last updated 1 minute ago

Actions Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associations
-	rtb-06a44a0c89fa60260	-
☒ Private Route Table	rtb-07e9b6a63180c7700	subnet-0bb16a3...
-	rtb-026c9b4827b268e0e	-

rtb-07e9b6a63180c7700 / Private Route Table

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated
0.0.0.0/0	nat-088095ca4b1237e26	Active	No
10.0.0.0/16	local	Active	No

Set main route table

Main route table controls the routing for all subnets that are not explicitly associated with any other route table. Are you sure you want to set this route table as the main route table?

- rtb-07e9b6a63180c7700 / Private Route Table

To confirm setting, type set in the field.

set

Cancel OK

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Click ok

Subnet association -> **Edit subnet associations**

VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#EditRouteTableSubnetAssociations:RouteTableId=rtb-07e9b6a63180c7700

Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Route tables > rtb-07e9b6a63180c7700 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Lab-subnet-private1-us-east...	subnet-0bb16a3b3688f1e0e	10.0.1.0/24	-	rtb-07e9b6a63180c7700 / Privat
☐	Public Subnet 2	subnet-035345b76468ca63b	10.0.2.0/24	-	rtb-07e9b6a63180c7700 / Privat
☐	Lab-subnet-public1-us-east...	subnet-01586705d9af70b06	10.0.0.0/24	-	rtb-07e9b6a63180c7700 / Privat
☒	Private Subnet 2	subnet-02cfe32afe753cac9	10.0.3.0/24	-	rtb-07e9b6a63180c7700 / Privat

Selected subnets

subnet-02cfe32afe753cac9 / Private Subnet 2 X subnet-0bb16a3b3688f1e0e / Lab-subnet-private1-us-east-2a X

Cancel Save associations

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Click save association

VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTables:

Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Route tables

Route tables (1/4) Info

Last updated 2 minutes ago Actions Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	M
☐	-	rtb-06a44a0c89fa60260	-	-	Ni
☐	Private Route Table	rtb-07e9b6a63180c7700	4 subnets	-	Ye
☐	-	rtb-026c9b4827b268e0e	-	-	Ye
☒	Lab-rtb-public	rtb-08ab4755b712ef034	-	-	Ni

Edit Name

Public Route Table

Cancel Save

rtb-08ab4755b712ef034 / Lab-rtb-public

Details Routes Subnet associations Edge associations Route propagation Tags

Details

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VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTables:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Route tables

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Route tables (1/4) Info

Last updated less than a minute ago

Actions Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	M
-	rtb-06a44a0c89fa60260	-	-	Ni
Private Route Table	rtb-07e9b6a63180c7700	4 subnets	-	Ye
-	rtb-026c9b4827b268e0e	-	-	Ye
Public Route Table	rtb-08ab4755b712ef034	-	-	Ni

Routes (2)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-041642af4f7ba4579	Active	No
10.0.0.0/16	local	Active	No

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VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTables:

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Route tables

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

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Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Route tables (1/4) Info

Last updated 1 minute ago

Actions Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	M
-	rtb-06a44a0c89fa60260	-	-	Ni
Private Route Table	rtb-07e9b6a63180c7700	4 subnets	-	Ye
-	rtb-026c9b4827b268e0e	-	-	Ye
Public Route Table	rtb-08ab4755b712ef034	-	-	Ni

rtb-08ab4755b712ef034 / Public Route Table

Details Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (0)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnet associations

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VPC | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#EditRouteTableSubnetAssociations:RouteTableId=rtb-08ab4755b712ef034

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52006044 @ 3379-6396-0076

VPC > Route tables > rtb-08ab4755b712ef034 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	Lab-subnet-private1-us-eas...	subnet-0bb16a3b3688f1e0e	10.0.1.0/24	-	rtb-07e9b6a63180c7700 / Privat
☒	Public Subnet 2	subnet-035345b76468ca63b	10.0.2.0/24	-	Main (rtb-07e9b6a63180c7700 /
☒	Lab-subnet-public1-us-east...	subnet-01586705d9af70b06	10.0.0.0/24	-	Main (rtb-07e9b6a63180c7700 /
☐	Private Subnet 2	subnet-02cfe32afe753cac9	10.0.3.0/24	-	rtb-07e9b6a63180c7700 / Privat

Selected subnets

subnet-035345b76468ca63b / Public Subnet 2 X subnet-01586705d9af70b06 / Lab-subnet-public1-us-east-2a X

Cancel Save associations

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labclient.labondemand.com/Instructions/ExamResult/0147cdef-d6cb-438d-8eae-f41533f9e4a6 - Work - Microsoft Edge

https://labclient.labondemand.com/Instructions/ExamResult/0147cdef-d6cb-438d-8eae-f41533f9e4a6

Congratulations!

Print

Enjoy the success you've worked so hard to achieve!



Areas where you excelled

- Create a base network by using the VPC Wizard**
You completed this section with ease.
- Create additional subnets**
Looks like you have a strong grasp on this material.
- Configure route tables**
You completed this area with great success.

94°F Haze 7:03 PM 6/7/2025

You are done with the lab!!!!